UNIT III

Biaxial Strength Theories & Laminate Analysis

Max stress | max strain | Tsai-Hill | Tsai-Wu | Kirchhoff hypothesis | CLT | A, B, D matrices

I. Biaxial Strength Theories of a Lamina

A lamina is usually under a **combined (biaxial) stress state** σ_1, σ_2, τ_{12} in its material axes. A **failure criterion** predicts whether the lamina fails, by comparing this state with the basic strengths obtained from simple tests. Because the lamina is **anisotropic**, strength depends on direction and on the sign of stress (tension ≠ compression).

I.1 The five basic strengths of a UD lamina

Symbol	Strength	Meaning
X_t (X)	Longitudinal tensile	along fibres, tension
X_c (X′)	Longitudinal compressive	along fibres, compression (microbuckling)
Y_t (Y)	Transverse tensile	⊥ fibres, tension (matrix / interface)
Y_c (Y′)	Transverse compressive	⊥ fibres, compression
S	In-plane shear	τ_{12} at failure

Material	X_t	X_c	Y_t	Y_c	S (MPa)
Glass/epoxy	1062	610	31	118	72
Carbon/epoxy (T300)	1500	1500	40	246	68
Kevlar/epoxy	1400	235	12	53	34

Note: $Y \ll X$: a lamina is **20–40 times weaker** across the fibres - the matrix cracks first (first-ply failure).

I.2 General procedure

- Find the applied stresses σ_x, σ_y, τ_{xy} (global axes).
- Transform to material axes: $\{\sigma_1, \sigma_2, \tau_{12}\}^T = [T]\{\sigma_x, \sigma_y, \tau_{xy}\}^T$.
- Apply the criterion using the five strengths; failure if the condition is met or exceeded.
- Strength ratio R = (failure stress)/(applied stress); **R > 1 safe**, R = 1 failure, R < 1 fails.

II. Maximum Stress Theory

Failure occurs when **any one** stress component in the material axes reaches its corresponding strength. The components act **independently** (no interaction).

$$-X_c < \sigma_1 < X_t \; ; \; -Y_c < \sigma_2 < Y_t \; ; \; |\tau_{12}| < S$$

If **any** inequality is violated the lamina fails, and the mode of failure is identified directly (fibre failure, matrix failure or shear).

2.1 Off-axis uniaxial tension x

For a lamina loaded by σ_x at angle θ ($\sigma_1 = \sigma_x \cos^2\theta$, $\sigma_2 = \sigma_x \sin^2\theta$, $\tau_{12} = -\sigma_x \sin\theta\cos\theta$), the strength is the **smallest** of:

$$\sigma_x = X_t \,/\, \cos^2\theta \text{ (fibre failure)}$$
$$\sigma_x = Y_t \,/\, \sin^2\theta \text{ (transverse failure)}$$
$$\sigma_x = S \,/\, (\sin\theta\cos\theta) \text{ (shear failure)}$$

- Predicts **abrupt kinks** in strength-vs-angle curve where the failure mode changes.
- Simple, mode-specific; but ignores **interaction** of stresses, so it is often unconservative near mixed states.

III. Maximum Strain Theory

Failure occurs when **any strain component** in the material axes reaches its ultimate value. It allows for **Poisson interaction** because strains depend on all stresses.

$$-\varepsilon_{1c} < \varepsilon_1 < \varepsilon_{1t} \; ; \; -\varepsilon_{2c} < \varepsilon_2 < \varepsilon_{2t} \; ; \; |\gamma_{12}| < \gamma_{12u}$$
$$\varepsilon_{1t} = X_t/E_1 \; , \; \varepsilon_{2t} = Y_t/E_2 \; , \; \gamma_{12u} = S/G_{12}$$

With plane stress: $\varepsilon_1 = (\sigma_1 - \nu_{12}\sigma_2)/E_1$, $\varepsilon_2 = (\sigma_2 - \nu_{21}\sigma_1)/E_2$, $\gamma_{12} = \tau_{12}/G_{12}$. For off-axis tension:

$$\sigma_x = X_t \,/\, (\cos^2\theta - \nu_{12} \sin^2\theta) \; , \; \sigma_x = Y_t \,/\, (\sin^2\theta - \nu_{21} \cos^2\theta) \; , \; \sigma_x = S/(\sin\theta \cos\theta)$$

Compare: Max stress and max strain coincide only when $\nu_{12} = 0$. Both are **non-interactive** criteria - each mode is checked separately.

IV. Tsai-Hill Theory (Interactive)

Hill (1948) extended the **von Mises** yield criterion to anisotropic materials; Tsai adapted it to composites. A **single quadratic equation** combines all stress components, so one stress lowers the allowable value of the others (**interaction**). For a plane-stress lamina:

$$\sigma_1^2 \,/\, X^2 - \sigma_1\sigma_2 \,/\, X^2 + \sigma_2^2 \,/\, Y^2 + \tau_{12}^2 \,/\, S^2 = 1$$

- Use $X = X_t$ if $\sigma_1 > 0$ else X_c; $Y = Y_t$ if $\sigma_2 > 0$ else Y_c (strengths chosen by sign of stress).
- Failure if LHS ≥ 1; **strength ratio** $R = 1/\sqrt{(LHS)}$.

- Does not distinguish failure mode; assumes equal tension and compression strengths in each direction.
- Off-axis tension: $1/\sigma_x^2 = \cos^4\theta/X^2 + (1/S^2 - 1/X^2)\sin^2\theta\cos^2\theta + \sin^4\theta/Y^2$ (smooth curve).

Origin of Tsai-Hill (Hill, 1948)

> Hill's anisotropic yield function: $F(\sigma_2 - \sigma_3)^2 + G(\sigma_3 - \sigma_1)^2 + H(\sigma_1 - \sigma_2)^2 + 2L\tau_{23}^2 + 2M\tau_{31}^2 + 2N\tau_{12}^2 = 1$.

> Apply to simple tests: $\sigma_1 = X$ alone → $G + H = 1/X^2$; $\sigma_2 = Y$ → $F + H = 1/Y^2$; $\sigma_3 = Z$ → $F + G = 1/Z^2$; $\tau_{12} = S$ → $2N = 1/S^2$.

> For a UD lamina (transversely isotropic, $Y = Z$, plane stress $\sigma_3 = 0$): $F + G$... gives $G = 1/(2X^2)\cdot(\ldots)$ and finally $H = 1/(2X^2)$, $G + H = 1/X^2$.

> Substituting → $\sigma_1^2/X^2 - \sigma_1\sigma_2/X^2 + \sigma_2^2/Y^2 + \tau_{12}^2/S^2 = 1$.

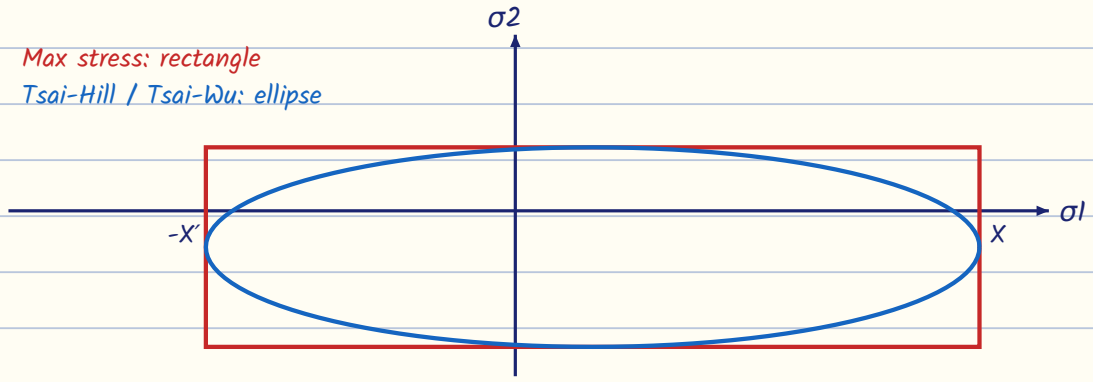


Fig. 1 Failure envelope in 1-2 space ($\tau_{12} = 0$): max stress vs Tsai-Hill / Tsai-Wu

V. Tsai-Wu Tensor Theory

Tsai and Wu (1971) proposed a **tensor polynomial** failure criterion that accounts for **different tensile and compressive strengths** and for interaction through a cross term F_{12}. It reduces to Tsai-Hill in a special case.

$$F_i\sigma_i + F_{ij}\sigma_i\sigma_j = 1 \quad (i, j = 1, 2, 6)$$

$$F_1\sigma_1 + F_2\sigma_2 + F_{11}\sigma_1^2 + F_{22}\sigma_2^2 + F_{66}\tau_{12}^2 + 2F_{12}\sigma_1\sigma_2 = 1$$

$$F_1 = 1/X_t - 1/X_c \,,\; F_2 = 1/Y_t - 1/Y_c$$

$$F_{11} = 1/(X_tX_c) \,,\; F_{22} = 1/(Y_tY_c) \,,\; F_{66} = 1/S^2$$

$$F_{12} = -\tfrac{1}{2}\sqrt{(F_{11}F_{22})} \text{ (empirical, from biaxial test)}$$

- F_i terms (linear) account for the **tension–compression strength difference**; F_{ij} terms are quadratic.
- F_{12} needs a biaxial test; the stability condition is $F_{11}F_{22} - F_{12}^2 > 0$, i.e. $-\sqrt{(F_{11}F_{22})} < F_{12} < \sqrt{(F_{11}F_{22})}$.

- Strength ratio R solves $aR^2 + bR - 1 = 0$: $a = F_{ij}\sigma_i\sigma_j$, $b = F_i\sigma_i \rightarrow R = [-b + \sqrt{(b^2 + 4a)}] / 2a$.
- Shear is **sign-insensitive** ($F_6 = 0$, $F_{16} = F_{26} = 0$ for orthotropic lamina).

Theory	Interaction	Tension ≠ compression	Failure mode	Remark
Max stress	No	Yes	Yes	Simple; unconservative
Max strain	Poisson only	Yes	Yes	Slightly different
Tsai-Hill	Yes	No (select by sign)	No	Smooth; good for UD
Tsai-Wu	Yes	Yes	No	Most general, widely used

VI. Numerical Problems on Strength Theories

Problem 1

Glass/epoxy lamina ($E_1 = 38.6$, $E_2 = 8.27$, $G_{12} = 4.14$ GPa, $\nu_{12} = 0.26$; $X_t = 1062$, $X_c = 610$, $Y_t = 31$, $Y_c = 118$, $S = 72$ MPa) carries $\sigma_1 = 500$, $\sigma_2 = 20$, $\tau_{12} = 30$ MPa. Check failure by all four theories.

(a) Max stress: $500 < 1062$ ✓ ; $20 < 31$ ✓ ; $30 < 72$ ✓ → safe. $R = \min(1062/500, 31/20, 72/30) = \min(2.12, 1.55, 2.40) = \mathbf{1.55}$ (transverse tension governs).

(b) Max strain: $\varepsilon_1 = (500 - 0.26 \times 20)/38600 = 0.01282 < \varepsilon_{1t} = 0.02751$ ✓ ; $\varepsilon_2 = (20 - 0.0557 \times 500)/8270 = -0.00095$ (compressive, limit 0.01427) ✓ ; $\gamma_{12} = 30/4140 = 0.00725 < 0.01739$ ✓ → safe.

(c) Tsai-Hill: $(500/1062)^2 - (500 \times 20)/1062^2 + (20/31)^2 + (30/72)^2 = 0.2217 - 0.0089 + 0.4162 + 0.1736 = \mathbf{0.803 < 1}$ → safe; $R = 1/\sqrt{0.803} = \mathbf{1.12}$.

(d) Tsai-Wu: $F_1 = -6.98 \times 10^{-4}$, $F_2 = 0.02378$, $F_{11} = 1.544 \times 10^{-6}$, $F_{22} = 2.734 \times 10^{-4}$, $F_{66} = 1.929 \times 10^{-4}$, $F_{12} = -1.027 \times 10^{-5}$. Sum $= -0.349 + 0.476 + 0.386 + 0.109 + 0.174 - 0.205 = \mathbf{0.590 < 1}$ → safe; $R = \mathbf{1.34}$.

Observe: Different theories give different margins (1.55 / 1.12 / 1.34); Tsai-Hill is most conservative here because it includes interaction.

Problem 2

The same lamina has fibres at $\theta = 30°$ to the loading axis. Find the maximum σ_x (uniaxial tension) by max stress and by Tsai-Hill.

$\cos^2 30° = 0.75$, $\sin^2 30° = 0.25$, $\sin\theta\cos\theta = 0.433$.

Max stress: $X/\cos^2 = 1062/0.75 = 1416$; $Y/\sin^2 = 31/0.25 = \mathbf{124}$; $S/(sc) = 72/0.433 = 166$ MPa → σ_x = **124 MPa** (matrix failure).

Tsai-Hill: $1/\sigma_x^2 = 0.5625/1062^2 + (1/72^2 - 1/1062^2)(0.1875) + 0.0625/31^2 = 1.0 \times 10^{-4}$ → σ_x = **99.2 MPa** (20 % lower).

VII. Macro Mechanical Analysis of a Laminate

A **laminate** is a stack of laminae with different fibre angles bonded together. **Classical Lamination Theory (CLT)** relates the **in-plane loads N** and **moments M** applied to the laminate to the **mid-plane strains** ε^0 and **curvatures** κ.

7.1 Laminate code (stacking sequence notation)

- Plies listed from the **bottom (or top) to the other face**, angle in degrees, separated by "/": **[0/45/-45/90]**.
- **Subscript s** = symmetric about the mid-plane: $[0/90]_s = [0/90/90/0]$. Subscript **T** = total: $[0/90]_T$ = two plies.
- Subscript number = repeat: $[0/90]_2 = [0/90/0/90]$; $[0/90]_{2s} = [0/90/0/90/90/0/90/0]$.
- **±45** = a +45° ply above a -45° ply ($[\pm 45] = [45/-45]$); an overbar / "o" marks the mid-ply of an odd-ply symmetric laminate: [0/90/0].
- Thicknesses are normally equal, so ply count × t gives total h.

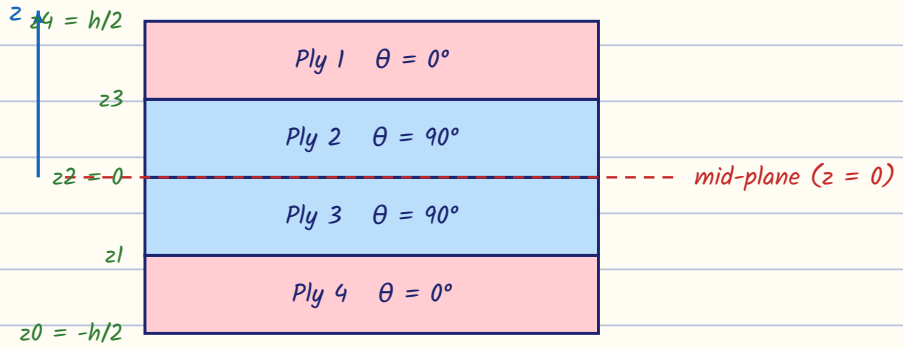


Fig. 2 Laminate [0/90]s with mid-plane and z-coordinates of ply surfaces

Type of laminate	Definition	Example
Symmetric	Plies and angle symmetric about mid-plane: B = 0	[0/45/90]s
Anti-symmetric	Equal ± angles at equal ±z, opposite sign	[45/-45/45/-45]
Balanced	For every +θ ply there is a -θ ply: $A_{16} = A_{26} = 0$	[30/-30]s
Cross-ply	Only 0° and 90° plies	[0/90/0]

Type of laminate	Definition	Example
Angle-ply	Only +θ and -θ plies	[±30]s
Quasi-isotropic	In-plane stiffness same in all directions	[0/±60]s , [0/±45/90]s

VIII. Kirchhoff Hypothesis (Classical Lamination Theory Assumptions)

- The laminate is **thin** ($h \ll$ in-plane dimensions), made of **perfectly bonded** laminae (no slip, bond infinitesimally thin).
- **Normals** to the mid-plane remain **straight** and **normal** to the deformed mid-plane (no transverse shear strains $\gamma_{xz} = \gamma_{yz} = 0$).
- Normals are **inextensible**: the thickness does not change ($\varepsilon_z = 0$).
- Displacements and strains are **small**; each lamina is linear elastic and in **plane stress** ($\sigma_z = \tau_{xz} = \tau_{yz} = 0$).

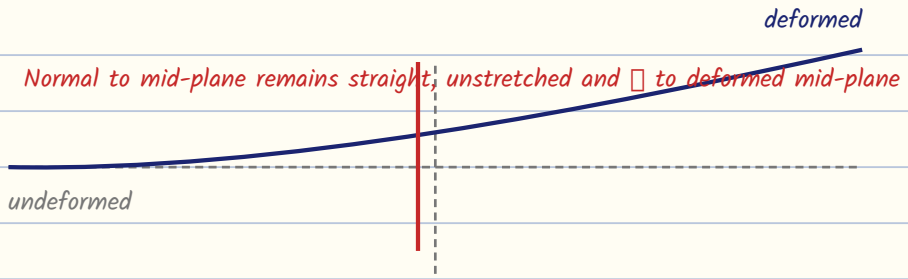


Fig. 3 Kirchhoff-Love kinematics: a normal remains straight and perpendicular after bending

8.1 Displacement and strain field

Let u^0, v^0, w be mid-plane displacements. A point at distance z from the mid-plane moves by:

$$u = u^0 - z\,\partial w/\partial x \;,\; v = v^0 - z\,\partial w/\partial y \;,\; w = w(x, y)$$

Strains from strain-displacement relations ($\varepsilon_x = \partial u/\partial x$, $\varepsilon_y = \partial v/\partial y$, $\gamma_{xy} = \partial u/\partial y + \partial v/\partial x$):

$$\{\varepsilon_x\ \varepsilon_y\ \gamma_{xy}\}^T = \{\varepsilon_x^0\ \varepsilon_y^0\ \gamma_{xy}^0\}^T + z\,\{\kappa_x\ \kappa_y\ \kappa_{xy}\}^T$$

$$\kappa_x = -\partial^2 w/\partial x^2 \;,\; \kappa_y = -\partial^2 w/\partial y^2 \;,\; \kappa_{xy} = -2\,\partial^2 w/\partial x \partial y$$

Strain varies **linearly through thickness** ($\varepsilon = \varepsilon^0 + z\kappa$) - but **stress does not**, since $\bar{Q}$ differs from ply to ply: stress is **piece-wise linear / discontinuous** at ply interfaces.

8.2 Stresses in the k-th ply

$$\{\sigma\}_k = [\bar{Q}]_k\,\{\varepsilon\} = [\bar{Q}]_k\,(\{\varepsilon^0\} + z\,\{\kappa\})$$

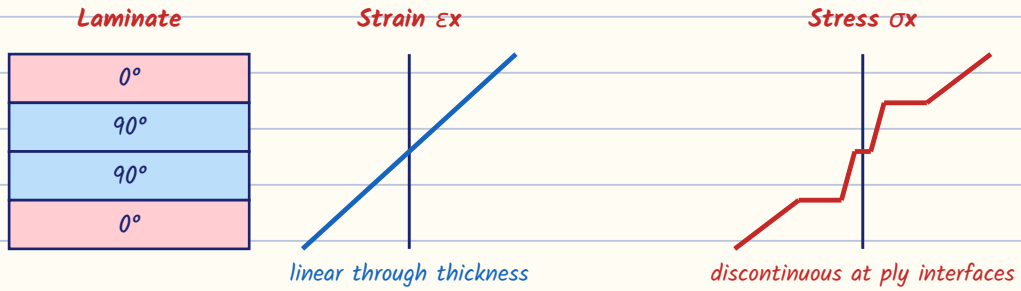


Fig. 4 [0/90]s in bending: linear strain but piece-wise (discontinuous) stress

IX. Force & Moment Resultants – A, B, D Matrices (Derivation)

Replace the stress distribution by **statically equivalent resultants** per unit width, obtained by integrating through the thickness h (z from -h/2 to +h/2):

$$\{N\} = \int \{\sigma\}\, dz \ (N/mm)\ ,\ \{M\} = \int \{\sigma\}\, z\, dz \ (N\cdot mm/mm)$$

$$N = (N_x, N_y, N_{xy})\ ,\ M = (M_x, M_y, M_{xy})$$

Because $\bar{Q}$ is constant within a ply, integrate **ply by ply** between z_{k-1} and z_k (N plies) and substitute $\{\sigma\}_k = [\bar{Q}]_k(\{\varepsilon^0\} + z\{\kappa\})$:

$$\{N\} = \Sigma_{k=1}^{n} \int_{z_{k-1}}^{z_k} [\bar{Q}]_k(\{\varepsilon^0\} + z\{\kappa\})\, dz$$

$$\{M\} = \Sigma_{k=1}^{n} \int_{z_{k-1}}^{z_k} [\bar{Q}]_k(\{\varepsilon^0\}\, z + z^2\{\kappa\})\, dz$$

Since ε^0 and κ are independent of z they come outside the summation; integrating gives:

$$\{N\} = [A]\{\varepsilon^0\} + [B]\{\kappa\}$$

$$\{M\} = [B]\{\varepsilon^0\} + [D]\{\kappa\}$$

$$A_{ij} = \Sigma\, \bar{Q}_{ij,k}\,(z_k - z_{k-1}) \ \text{(extensional stiffness)}$$

$$B_{ij} = \tfrac{1}{2}\,\Sigma\, \bar{Q}_{ij,k}\,(z_k^{\,2} - z_{k-1}^{\,2}) \ \text{(coupling stiffness)}$$

$$D_{ij} = \tfrac{1}{3}\,\Sigma\, \bar{Q}_{ij,k}\,(z_k^{\,3} - z_{k-1}^{\,3}) \ \text{(bending stiffness)}$$

$$\begin{bmatrix} A11 & A12 & A16 & B11 & B12 & B16 \\ A12 & A22 & A26 & B12 & B22 & B26 \\ A16 & A26 & A66 & B16 & B26 & B66 \\ B11 & B12 & B16 & D11 & D12 & D16 \\ B12 & B22 & B26 & D12 & D22 & D26 \\ B16 & B26 & B66 & D16 & D26 & D66 \end{bmatrix}$$

The 6 × 6 laminate stiffness matrix: {N, M} = [[A, B],[B, D]] {, }

Physical meaning

> **[A]** (kN/mm): in-plane (extensional) stiffness - relates N to ε^0. A_{16}, A_{26} = shear-extension coupling.

> **[D]** (kN·mm): bending / twisting stiffness - relates M to κ; D_{16}, D_{26} = bend-twist coupling. $D \propto h^3$ - outer plies dominate.

> **[B]**: **bending-extension coupling**; non-zero only for **unsymmetric** laminates: in-plane load causes bending/warping, and moment causes mid-plane strain.

> Units: A = GPa·mm, B = GPa·mm², D = GPa·mm³.

9.1 Inverted (compliance) form and ply stresses

$$\{\varepsilon^0\} = [a]\{N\} + [b]\{M\}\ ,\ \{\kappa\} = [c]\{N\} + [d]\{M\}$$

[a], [b], [c], [d] from inverting the 6×6 matrix ; for symmetric laminate: $\{\varepsilon^0\} = [A]^{-1}\{N\}$, $\{\kappa\} = [D]^{-1}\{M\}$

Procedure for laminate analysis

1. Get ply properties → [Q] → $[\bar{Q}]_k$ for each ply angle θ_k.
2. Locate z_k from the mid-plane: $z_0 = -h/2$, $z_k = z_{k-1} + t_k$.
3. Compute [A], [B], [D] by the summations above.
4. Solve for $\{\varepsilon^0\}$, $\{\kappa\}$ for given {N}, {M}.
5. Strain in each ply: $\varepsilon = \varepsilon^0 + z\kappa$; stress $\{\sigma\}_{xy} = [\bar{Q}]\{\varepsilon\}$; transform to material axes $\{\sigma\}_{12} = [T]\{\sigma\}_{xy}$.
6. Apply a failure criterion (Tsai-Wu etc.) → **first ply failure (FPF)**, then last ply failure with degraded stiffness.

X. Special Cases of Laminates

Laminate	Effect on [A], [B], [D]	Consequence
Symmetric (mirror about mid-plane)	[B] = 0 ($B_{ij} = 0$ for all i, j)	No bending-stretching coupling; no warping on cure; {N} and {M} decoupled
Balanced (+θ and -θ plies)	$A_{16} = A_{26} = 0$	No shear-extension coupling in plane
Cross-ply (0°/90° only)	$A_{16} = A_{26} = 0$, $D_{16} = D_{26} = 0$, $B_{16} = B_{26} = B_{66} = 0$	Specially orthotropic; unsymmetric cross-ply [0/90] has $B_{11} = -B_{22} \neq 0$ → curved on cooling
Angle-ply $[\pm\theta]_s$	$A_{16} = A_{26} = 0$ but D_{16}, $D_{26} \neq 0$	Bend-twist coupling remains

Laminate	Effect on [A], [B], [D]	Consequence
Quasi-isotropic $[0/\pm60]_s$, $[0/\pm45/90]_s$	$A_{11} = A_{22}$, $A_{66} = (A_{11} - A_{12})/2$, $B = 0$	In-plane isotropic (D is not)
Single isotropic layer	$A = Eh/(1-\nu^2) \cdot [1\ \nu;\ \nu\ 1]$, $B = 0$, $D = Eh^3/12(1-\nu^2)$	Reduces to classical plate theory

Design rule: Use **symmetric & balanced** laminates (e.g. $[0/\pm45/90]_s$) to avoid warping and coupling; place 0° plies outward for bending stiffness.

XI. Numerical Problems on Laminates

Problem 3

Glass/epoxy plies ($E_1 = 38.6$, $E_2 = 8.27$, $G_{12} = 4.14$ GPa, $\nu_{12} = 0.26$), $t = 0.125$ mm each. Find [A], [B], [D] for the laminate $[0/90]_s$.

$Q = [39.17\ 2.18\ 0\ ;\ 2.18\ 8.39\ 0\ ;\ 0\ 0\ 4.14]$ GPa for 0°; for 90°: $Q_{11} = 8.39$, $Q_{22} = 39.17$, $Q_{12} = 2.18$, $Q_{66} = 4.14$. $h = 4 \times 0.125 = 0.5$ mm; $z = -0.25, -0.125, 0, 0.125, 0.25$.

$A_{11} = (2 \times 39.17 + 2 \times 8.39)(0.125) =$ **11.89** ; $A_{22} =$ **11.89** ; $A_{12} = 4 \times 2.18 \times 0.125 =$ **1.09** ; $A_{66} = 4 \times 4.14 \times 0.125 =$ **2.07** (GPa·mm). $A_{16} = A_{26} = 0$.

$[B] =$ **0** (symmetric laminate). $D_{11} =$ **0.368** , $D_{22} =$ **0.1275** , $D_{12} =$ **0.0227** , $D_{66} =$ **0.0431** (GPa·mm³).

Check: $D_{11} \gg D_{22}$ because the 0° plies are at the **outside** (large z) - bending stiffness along x is 3 times that along y.

Problem 4

Same plies. For $[0/90]_s$ under $N_x = 50$ N/mm (others zero), find mid-plane strains, ply stresses and first-ply failure ($Y_t = 31$ MPa).

$[A] = [11.89\ 1.09\ ;\ 1.09\ 11.89]$ kN/mm → $\varepsilon_x^0 = A_{22}N_x/(A_{11}A_{22} - A_{12}^2) =$ **4.24×10^{-3}** ; $\varepsilon_y^0 = -A_{12}N_x/(\dots) =$ **-3.89×10^{-4}** ; effective $E_x = 23.6$ GPa.

0° plies: $\sigma_1 = Q_{11}\varepsilon_x + Q_{12}\varepsilon_y =$ **165.3 MPa** (along fibres, safe vs 1062).

90° plies (material axes): $\sigma_1 =$ **−6.0 MPa**, $\sigma_2 = Q_{12}\varepsilon_y + Q_{22}\varepsilon_x =$ **34.7 MPa > Y_t = 31** → transverse cracking of 90° plies.

FPF load = 50 × 31/34.7 = **44.6 N/mm** (laminate stress 89 MPa); the 0° plies still carry load → last-ply failure is much higher.

Problem 7

Apply Tsai-Wu to the 90° plies of Problem 4 ($\sigma_1 = -6.0$, $\sigma_2 = 34.7$ MPa) to find the strength ratio.

$a = F_{11}\sigma_1^2 + F_{22}\sigma_2^2 + 2F_{12}\sigma_1\sigma_2 = 0.334$; $b = F_1\sigma_1 + F_2\sigma_2 = 0.830$; $a + b = \mathbf{1.165 > 1}$ → fails.

$R = [-b + \sqrt{(b^2 + 4a)}]/2a = [-0.830 + \sqrt{(0.689 + 1.337)}]/0.668 = \mathbf{0.887}$ → FPF at $N_x = 50 \times 0.887 = \mathbf{44.4\ N/mm}$ (agrees with max-stress 44.6).

Problem 5

Find [A], [B] for the unsymmetric cross-ply [0/90] (two plies, t = 0.125 mm).

$A_{11} = (39.17 + 8.39)(0.125) = \mathbf{5.945}$, $A_{12} = \mathbf{0.545}$, $A_{66} = \mathbf{1.035}$ GPa·mm. $B_{11} = ½[39.17(0 - 0.125^2) + 8.39(0.125^2 - 0)] = \mathbf{-0.240}$, $B_{22} = \mathbf{+0.240}$ GPa·mm².

$\mathbf{B \neq 0}$ → a pure in-plane load N_x bends the plate; after cure cooling the plate **warps** (saddle shape).

Problem 6

Angle-ply $[45/-45]_s$, same plies: find A.

$A_{11} = A_{22} = \mathbf{8.56}$; $A_{12} = \mathbf{4.42}$; $A_{66} = \mathbf{5.40}$ GPa·mm ; $A_{16} = A_{26} = 0$ (balanced). B = 0; $D_{16} = D_{26} = \mathbf{0.060} \neq 0$ (bend-twist coupling).

Problem 8

$[0/90]_s$ glass/epoxy (t = 0.125 mm, D from Problem 3) is loaded by $M_x = 1$ N·mm/mm only. Find the curvatures and the extreme-fibre stress.

[D] = [0.3679 0.0227 ; 0.0227 0.1275] kN·mm, $M_x = 0.001$ kN·mm/mm (symmetric: B = 0 → N = 0 ⇒ $\varepsilon^0 = 0$).

det D = 0.3679 × 0.1275 - 0.0227² = 0.04640 ; $\kappa_x = D_{22}M_x/\text{det} = \mathbf{2.748 \times 10^{-3}\ /mm}$, $\kappa_y = -D_{12}M_x/\text{det} = \mathbf{-4.89 \times 10^{-4}\ /mm}$ (anticlastic).

At the outer face z = 0.25 mm: $\varepsilon_x = z\kappa_x = 6.87 \times 10^{-4}$, $\varepsilon_y = -1.22 \times 10^{-4}$ → 0° ply $\sigma_1 = Q_{11}\varepsilon_x + Q_{12}\varepsilon_y = \mathbf{26.6\ MPa}$; at the interface z = 0.125: **13.3 MPa**.

11.1 First-ply failure and last-ply failure

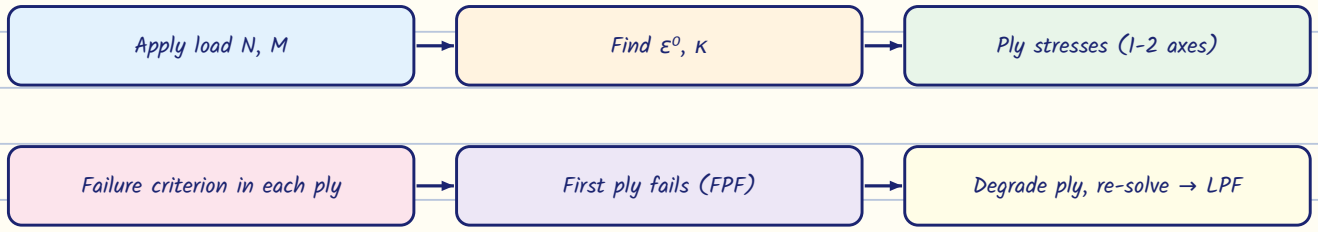


Fig. 5 Progressive failure analysis of a laminate

- **FPF**: load at which the first lamina (usually the 90° / off-axis ply by matrix cracking) satisfies the failure criterion.
- **LPF**: ultimate load after failed plies are discounted (E_2, G_{12}, $\nu_{12} \to 0$) and the load is re-distributed until fibre failure; LPF can be 2-3 times FPF.

XII. Quick Revision - Unit III

Remember

> Max stress / max strain: non-interactive; Tsai-Hill: interactive quadratic; Tsai-Wu: tensor polynomial with F_{12}, handles $X_t \neq X_c$.

> Kirchhoff: normals straight, normal, inextensible → $\varepsilon = \varepsilon^0 + z\kappa$. $N = A\varepsilon^0 + B\kappa$, $M = B\varepsilon^0 + D\kappa$.

> Symmetric → $B = 0$; balanced → $A_{16} = A_{26} = 0$; cross-ply → no coupling terms 16, 26.

Likely exam questions

> Explain max stress, max strain, Tsai-Hill and Tsai-Wu theories; compare them.

> State Kirchhoff hypothesis; derive A, B, D matrices in detail.

> Explain special cases of laminates; numerical on A, B, D and first-ply failure.

Short Answer Questions (2 marks)

Q1. List the five basic strengths of a lamina.

Ans: X_t, X_c, Y_t, Y_c, S.

Q2. Which theory is non-interactive?

Ans: Maximum stress and maximum strain theories.

Q3. Write Tsai-Hill criterion for plane stress.

Ans: $\sigma_1^2/X^2 - \sigma_1\sigma_2/X^2 + \sigma_2^2/Y^2 + \tau_{12}^2/S^2 = 1$.

Q4. Why is the F_{12} term required in Tsai-Wu?

Ans: It accounts for interaction between σ_1 and σ_2 (biaxial); it must be found from a biaxial test or approximated as $-\frac{1}{2}\sqrt{(F_{11}F_{22})}$.

Q5. State Kirchhoff's hypothesis.

Ans: Normals to the mid-plane stay straight, normal to the deformed mid-plane and unchanged in length.

Q6. What does [B] represent?

Ans: Bending-extension coupling stiffness; zero for symmetric laminates.

Q7. When are A_{16} and A_{26} zero?

Ans: For balanced laminates (equal $+\theta$ and $-\theta$ plies) and cross-ply laminates.

Q8. What is a quasi-isotropic laminate?

Ans: A laminate (e.g. $[0/\pm 60]_s$) whose in-plane stiffness is the same in all directions.

Q9. Define first-ply failure.

Ans: Load at which the first ply of the laminate satisfies a failure criterion.

Practice Problems with Answers

Problem A

Glass/epoxy lamina: $\sigma_1 = 300$, $\sigma_2 = -50$, $\tau_{12} = 40$ MPa. Use Tsai-Hill with strengths $X_t = 1062$, $Y_c = 118$, $S = 72$.

$(300/1062)^2 - 300(-50)/1062^2 + (50/118)^2 + (40/72)^2 = 0.0798 + 0.0133 + 0.1796 + 0.3086 = 0.581 < 1$ → safe, $R = 1.31$.

Problem B

Check max-stress failure of a lamina with $\sigma_1 = 1200$, $\sigma_2 = 10$, $\tau_{12} = 20$ MPa (same material).

$\sigma_1 = 1200 > X_t = 1062$ → **fibre failure** ($R = 0.885$).